Arunkarthick Ramesh

Mechanical Design Engineer — steel structures and lifting equipment

Lebenslauf · Karur, India



PERSÖNLICHE DATEN / Personal details

Wohnort	Karur, India · work base Chennai
Telefon	+91 98943 28064
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Nationalität	Indian
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Umzug	Willing to relocate

PROFIL

I design load-bearing steel and sheet-metal hardware, and I close it with a calculation. For eight years that has meant a Creo model, a GD&T drawing, and a strength check by hand and in Ansys, against EN 81-20 and EN 1993. The parts are elevator shaft components and steel brackets: they must carry load, be weldable, and be ready for production. On the critical jobs I do both the design and the structural check.

WAS ICH EINBRINGE / What I can do for the team

- Turn a requirement and a standard into a concept, a detailed 3D model in Creo, and a manufacturing drawing with GD&T.
- Check steel members, welds, and bolts to EN 1993 myself: hand calculation plus Ansys static, modal, and buckling, including the EN 1993-1-8 combined tension and shear check.
- Check brackets and lift frames as loaded steel structures: model, drawing, and strength, then release to production.
- Write the strength note, material callout, and release pack, and stay with the job through PDM. I lead six strength engineers who work this way.
- Answer production from the drawing: progressive, compound, and crank-press tools, welding fixtures, and drilling and boring jigs, from three years in a chain factory.
- Use Siemens NX on the project if that is the office system. I learned NX and SolidWorks in my degree. Creo and AutoCAD are what I use in industry.

BERUFSERFAHRUNG / Professional experience

Special Engineering Specialist

`12.2021 – heute · Chennai`

KONE Elevators · Shaft Mechanical & Component Engineering · Ace Performer 2023 and 2025

- Design and validate EN 81-20 load-bearing shaft components in Creo for European and Chinese elevators, about three projects a month: model, interference check, drawing.
- Check non-standard parts and large assemblies in Ansys Workbench and by manual EN 1993 calculation for the member, the weld, and the bolt. On one or two critical projects a quarter I do both the design and the analysis.
- Two months on site in Finland on high-rise elevator design and structural analysis with the European engineering team.
- Lead a six-person strength team from the input data to PDM release. Use 5 Whys when a design fault comes back. Report status to senior management and follow each job to the agreed delivery date.
- Six Sigma Green Belt initiative, still running: engineering time from 50 hours to 3 hours per project and cost from about €1,000 to €120, by widening bracket strength and the functional range, plus a further 30% cost reduction. With R&D, analysis time from 18 hours to 8 hours. A Power Apps register of past errors raised team efficiency 20% and cut repeat issues 5%.
- Ansys Mechanical 2025 R1, IronPython: read beam probes and joint probes, then check slip and EN 1993-1-8 tension, shear, and combined utilization. Bolt size comes from the beam radius. The script writes the legend and the PDF into the strength file.

Design Engineer

04.2019 – 12.2021 · Dindigul

Renold Chain India · Renold Einbeck, Germany

- For Renold Einbeck, 20–25 manufacturing drawings a month in AutoCAD and Creo, released in Windchill: new parts, AutoCAD-to-Creo conversion, and revisions. About four customer-approval and manufacturing drawings a month for Renold Chain India.
- Designed progressive, compound, and crank-press tools, welding fixtures, and drilling and boring jigs for special attachment chains, with the in-house manufacturing team.
- Modelled segmented and hub sprockets for the drive. Prepared CSR, BOM, and EDS, and applied heat treatment and metallurgy on the India drawing set. Standards: ISO 606 A (LH) and B (LL), ASME B29.8, BS 4116.
- Joined the global standardisation work: manufacturing drawings for chains and forming tools, plus BOM and EDS.

Business Development Engineer

01.2019 – 04.2019 · Dindigul

Renold Chain India

- Looked for new applications and segments, and ran CRM with OEMs, dealers, and the regional sales teams. Customer base up 20%.

Graduate Engineer Trainee

07.2018 – 12.2018 · Dindigul

Renold Chain India

- Time studies on hydraulic-press work, and manufacturing drawings for punch and tool parts. Pneumatic tools used on drive-chain assembly.

AUSBILDUNG / Education

B.E. Mechanical Engineering · CGPA 7.5 / 10

2014 – 2018 · Coimbatore

Sri Ramakrishna Institute of Technology, Anna University

- Academic CAD only, not used in my KONE or Renold work: Siemens NX and SolidWorks, for parts, assemblies, and drawings.
- Member, SAE. Member, The Institute of Indian Foundrymen.

KENNTNISSE / Skills

Industrie-CAD	Creo 9 (sheet metal, assemblies, Pro/Program), AutoCAD
Studium-CAD	Siemens NX, SolidWorks — degree only, not used at KONE or Renold
FEA	Ansys Workbench: static, modal, buckling, multi-load cases, mesh convergence. Explicit dynamics as a studied drop-test case. HyperMesh. Weld fatigue assessment.
Normen	EN 81-20, EN 1993 members, bolts, welds, and buckling, beam reactions for isolation selection, ISO 606 A (LH) and B (LL), ASME B29.8, BS 4116
Fertigung	Progressive and compound press tools, drill jigs, boring and welding fixtures, pneumatics on chain assembly
Freigabe	GD&T, Windchill / PDM, MathCAD, Power Apps, VBA and macros, strength reports, BOM, EDS

SPRACHEN / Languages

Englisch	Professional working language
Tamil	Native
Deutsch	A1, learning

ZERTIFIKATE / Certificates

- Six Sigma Yellow Belt — KONE. The Green Belt initiative above was a project, not this certificate.
- HyperMesh for FEA, plastics and sheet metal — Skill-Lync, May 2023.
- Geometric Dimensioning & Tolerancing — Udemy, May 2020.
- Sheet Metal Design — Udemy, March 2020.
- Innovation Award 2026, top 10 of 169 ideas. Ace Performer, KONE, 2023 and 2025.